# Sets and Sample Spaces

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MATH 241

Lecture #2

# Reminders!

- Homework #0 is due on Friday, 9/4 at 11:59pm
  - Includes survey, setup of tech, and turning in quick problem
- Read through Chapter 1 of Introduction to Probability
- Office hours: none on Monday, Tuesday from 3-4:30pm in Rocky 106A

# Learning Objectives

- **Familiarize** ourselves with the purpose of events and sample spaces
- **Visualize** subsets of the sample space when intersections, unions, and complements of event(s) are used
- **Introduce** techniques for proving theorems in general with the use of the proof of DeMorgan's laws

# Sets

*A set is a Many that allows itself to be thought of as a One.*

-Georg Cantor

# Sets

- A collection of objects
- Denoted as S
- $x \in S$ if it is an element of the set
    - How would the notation change if $x$ was not an element of $S$?

# Sets

Set of all even numbers $\rightarrow \{0, 2, 4, 6, 8, ...\}$

Set of places Jericho has lived in $\rightarrow$ {Phoenix, Riverside, ...}

All real numbers between 2 and 8 $\rightarrow [2, 8]$

Real coordinates within one unit of center $\rightarrow \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq 1\}$

Set of all outcomes when rolling 2 dice $\rightarrow$ ?

Set of all weather events that could occur today $\rightarrow$ ?

# Other Set-Related Items

- Empty set: $\{\}, \emptyset$
  - Denotes nothing existing inside
- Power set: $\mathcal{P}(S)$
  - Set of all subsets of $S$
- Subset: $A \subseteq B$
  - Every element of A must be an element of B
- Partition: $A_1 \cup A_2 \cup ... \cup A_n = S$
  - Collection of disjoint subsets; union is entire set
- Cardinality: $|A|$
  - Size; number of elements in set

# Sample Spaces

**Definition**

The sample space $S$ of an experiment is the set of all possible outcomes of the experiment.

# Examples

Birthdays of $n$ people in a room $\rightarrow$ S has $365^n$ outcomes*

Outcomes of two dice rolled $\rightarrow$ S has 36 outcomes

Infection status of two people $\rightarrow$ S has 4 outcomes

# Example of Two Dice Rolled

| | 2nd Dice | | | | | |
|---|---|---|---|---|---|---|
| 1st Dice | (1, 1) | (1, 2) | (1, 3) | (1, 4) | (1, 5) | (1, 6) |
| | (2, 1) | (2, 2) | (2, 3) | (2, 4) | (2, 5) | (2, 6) |
| | (3, 1) | (3, 2) | (3, 3) | (3, 4) | (3, 5) | (3, 6) |
| | (4, 1) | (4, 2) | (4, 3) | (4, 4) | (4, 5) | (4, 6) |
| | (5, 1) | (5, 2) | (5, 3) | (5, 4) | (5, 5) | (5, 6) |
| | (6, 1) | (6, 2) | (6, 3) | (6, 4) | (6, 5) | (6, 6) |

- $S = \{(1,1),(1,2),...,(6,6)\}$
- $S = \{(i,j) \in \mathbb{N}^2 : 1 \leq i \leq 6, 1 \leq j \leq 6\}$
- $|S| = 36$

# Event

## Definition

An event $E$ is any subset of the sample space; that is, it's a set consisting of possible outcomes of the sample space. If an outcome in $E$ has occurred, then we say the event $E$ has occurred.

# Examples

- Birthdays of $n$ people in a room
  - $E =$ at least two people that share the same birthday
- Outcomes of two dice rolled
  - $E =$ sum is odd
- Infection status of two people
  - $E =$ both people are infected

# Example of Two Dice Rolled

- $E =$ sum is odd
- $F =$ sum is $5$

# Example of Two Dice Rolled

- $E$ = sum is odd
- $F$ = sum is 5

| | 2nd Dice | | | | | |
|---|---|---|---|---|---|---|
| 1st Dice | (1, 1) | (1, 2) | (1, 3) | (1, 4) | (1, 5) | (1, 6) |
| | (2, 1) | (2, 2) | (2, 3) | (2, 4) | (2, 5) | (2, 6) |
| | (3, 1) | (3, 2) | (3, 3) | (3, 4) | (3, 5) | (3, 6) |
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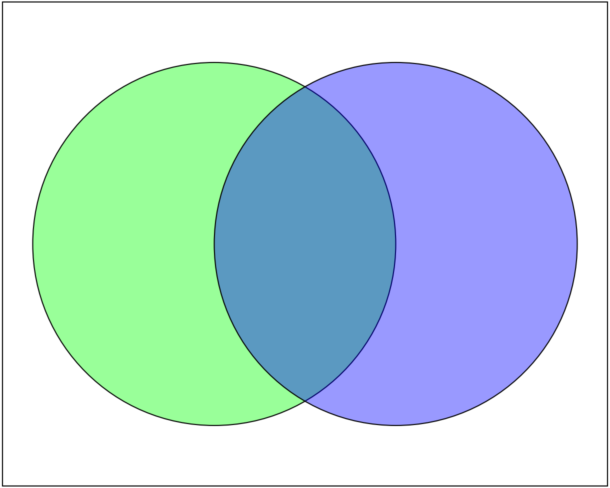
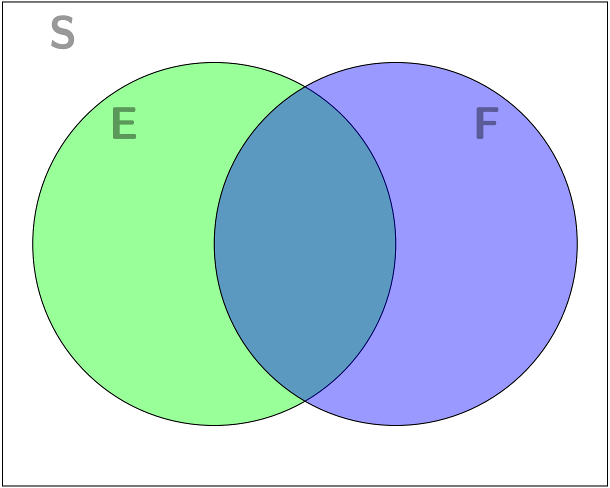
# Example of Two Dice Rolled

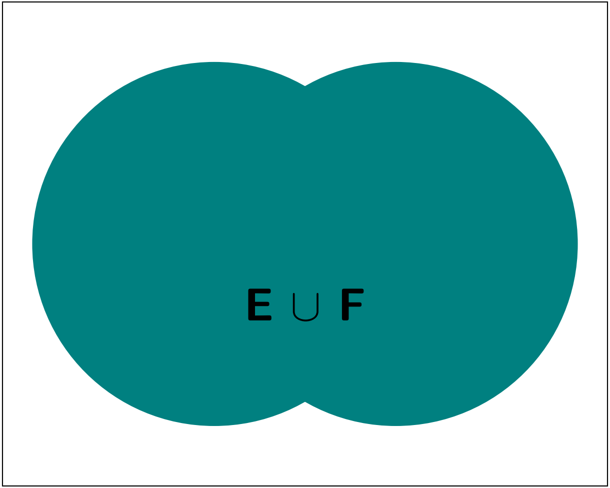
- $E$ = sum is odd
- $F$ = sum is 5

| | 2nd Dice | | | | | |
|---|---|---|---|---|---|---|
| | (1, 1) | (1, 2) | (1, 3) | (1, 4) | (1, 5) | (1, 6) |
| | (2, 1) | (2, 2) | (2, 3) | (2, 4) | (2, 5) | (2, 6) |
| 1st | (3, 1) | (3, 2) | (3, 3) | (3, 4) | (3, 5) | (3, 6) |
| Dice | (4, 1) | (4, 2) | (4, 3) | (4, 4) | (4, 5) | (4, 6) |
| | (5, 1) | (5, 2) | (5, 3) | (5, 4) | (5, 5) | (5, 6) |
| | (6, 1) | (6, 2) | (6, 3) | (6, 4) | (6, 5) | (6, 6) |

**Wait a minute?!** Elements of the sample space exist in both events. What does this mean??

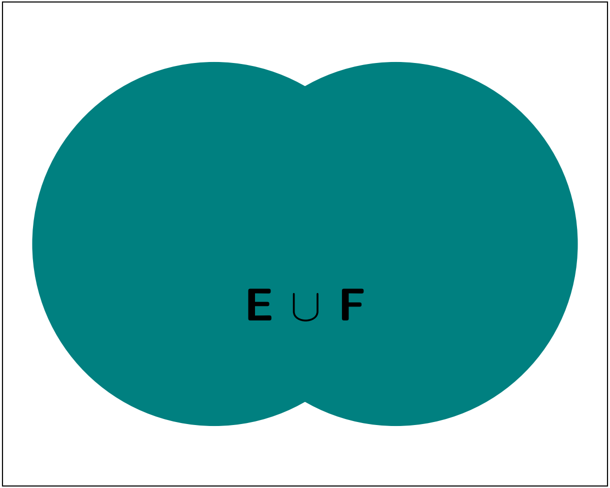



S
E
F




## Definition

The union of two events $E$ and $F$, which we write as $E \cup F$, is the event containing all outcomes in $E$ or $F$, i.e. $E \cup F = \{x | x \in E \text{ or } x \in F\}$.




- Inclusive “or”
- An element in at least one event

# Example of Two Dice Rolled

- $E =$ sum is odd
- $F =$ sum is 5
- $E \cup F =$ sum is **odd** or **5**

# Example of Two Dice Rolled

- $E =$ sum is odd
- $F =$ sum is 5
- $E \cup F =$ sum is **odd** or **5**

| | 2nd Dice | | | | | |
|---|---|---|---|---|---|---|
| | (1, 1) | **(1, 2)** | (1, 3) | **(1, 4)** | (1, 5) | **(1, 6)** |
| | **(2, 1)** | (2, 2) | **(2, 3)** | (2, 4) | **(2, 5)** | (2, 6) |
| 1st | (3, 1) | **(3, 2)** | (3, 3) | **(3, 4)** | (3, 5) | **(3, 6)** |
| Dice | **(4, 1)** | (4, 2) | **(4, 3)** | (4, 4) | **(4, 5)** | (4, 6) |
| | (5, 1) | **(5, 2)** | (5, 3) | **(5, 4)** | (5, 5) | **(5, 6)** |
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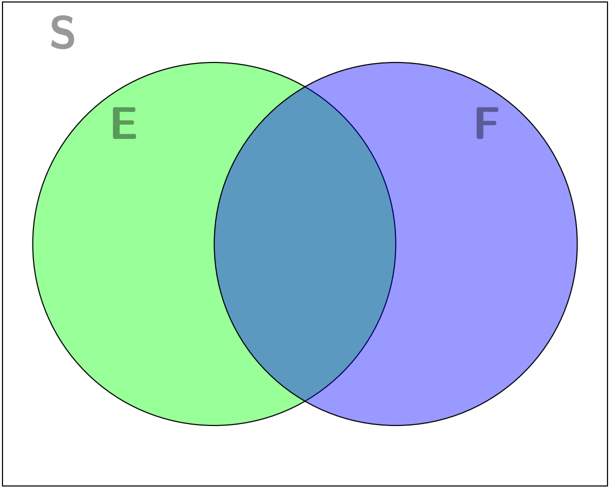
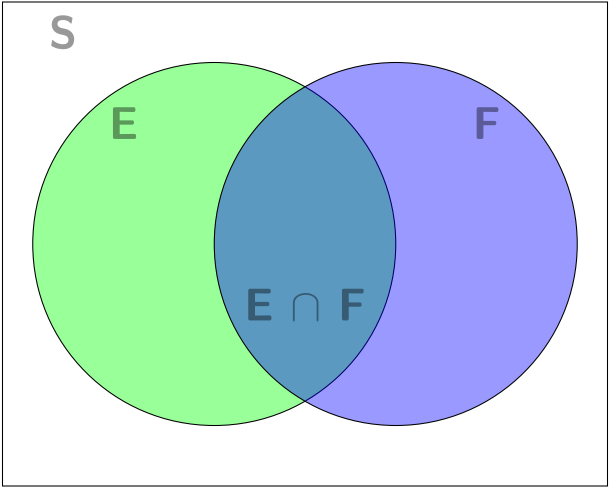
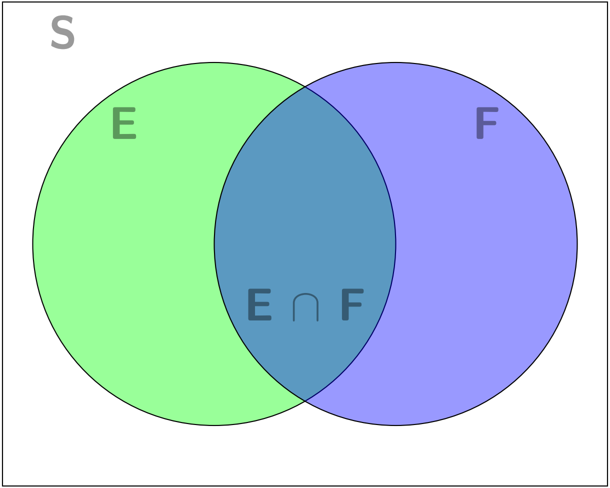

S
E
F




## Definition

The intersection of two events $E$ and $F$, which we write as $E \cap F$, is the event containing all outcomes in $E$ and $F$, i.e.
$E \cap F = \{x | x \in E \text{ and } x \in F\}$.




- "and"
- An element in all events

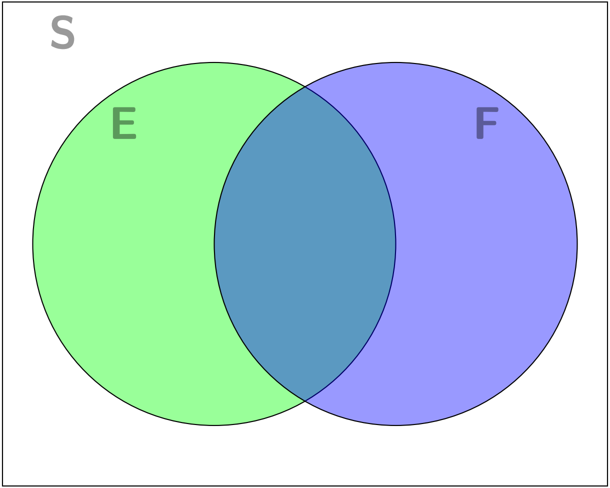
# Example of Two Dice Rolled

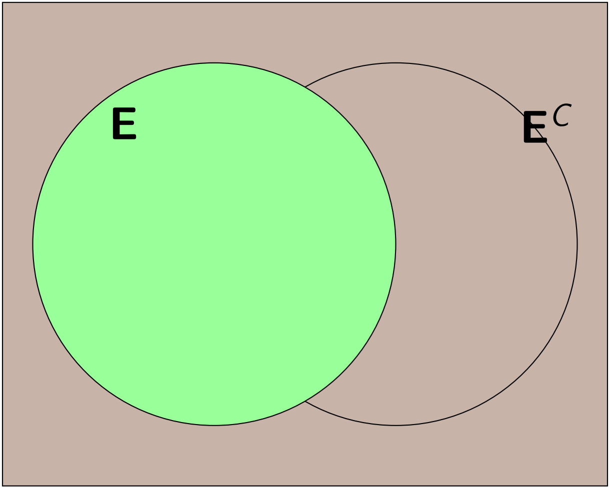
- $E =$ sum is odd
- $F =$ sum is 5
- $E \cap F =$ sum is **odd and 5**

# Example of Two Dice Rolled

- $E =$ sum is odd
- $F =$ sum is 5
- $E \cap F =$ sum is **odd and 5**

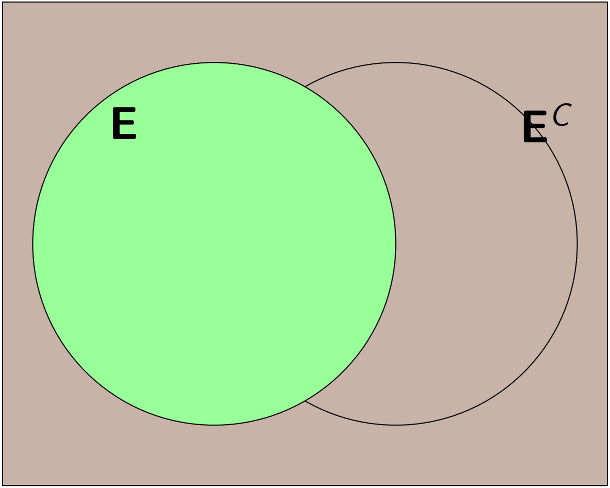
| | 2nd Dice | | | | | |
|---|---|---|---|---|---|---|
| | (1, 1) | (1, 2) | (1, 3) | **(1, 4)** | (1, 5) | (1, 6) |
| | (2, 1) | (2, 2) | **(2, 3)** | (2, 4) | (2, 5) | (2, 6) |
| 1st | (3, 1) | **(3, 2)** | (3, 3) | (3, 4) | (3, 5) | (3, 6) |
| Dice | **(4, 1)** | (4, 2) | (4, 3) | (4, 4) | (4, 5) | (4, 6) |
| | (5, 1) | (5, 2) | (5, 3) | (5, 4) | (5, 5) | (5, 6) |
| | (6, 1) | (6, 2) | (6, 3) | (6, 4) | (6, 5) | (6, 6) |


S
E
F




## Definition

The complement of $E$, which we write as $E^C$, is the event containing all outcomes in the sample space that are not in $E$, i.e. $E^C = \{x | x \notin E\}$.




- "not"
- $\backslash$ can be used to remove from an event (i.e. $E \backslash F$)
- Useful in situation where studying event $E$ is more difficult

# Example of Two Dice Rolled

- $E =$ sum is odd
- $F =$ sum is 5
- $E \notin S =$ sum is not **odd**

# Example of Two Dice Rolled

- $E$ = sum is odd
- $F$ = sum is 5
- $E \notin S$ = sum is not **odd**

| | 2nd Dice | | | | | |
|---|---|---|---|---|---|---|
| | **(1, 1)** | (1, 2) | **(1, 3)** | (1, 4) | **(1, 5)** | (1, 6) |
| | (2, 1) | **(2, 2)** | (2, 3) | **(2, 4)** | (2, 5) | **(2, 6)** |
| 1st | **(3, 1)** | (3, 2) | **(3, 3)** | (3, 4) | **(3, 5)** | (3, 6) |
| Dice | (4, 1) | **(4, 2)** | (4, 3) | **(4, 4)** | (4, 5) | **(4, 6)** |
| | **(5, 1)** | (5, 2) | **(5, 3)** | (5, 4) | **(5, 5)** | (5, 6) |
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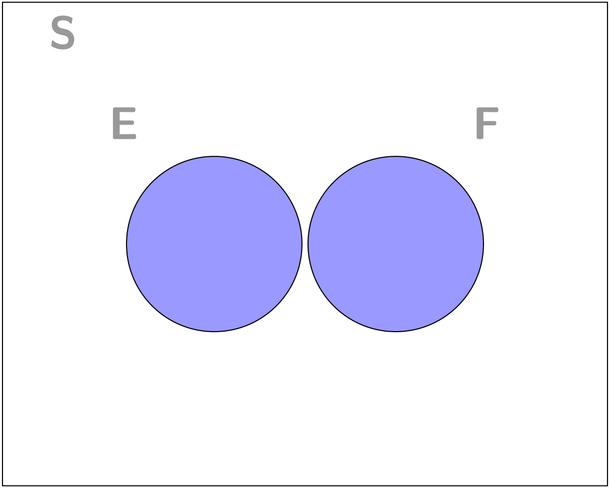
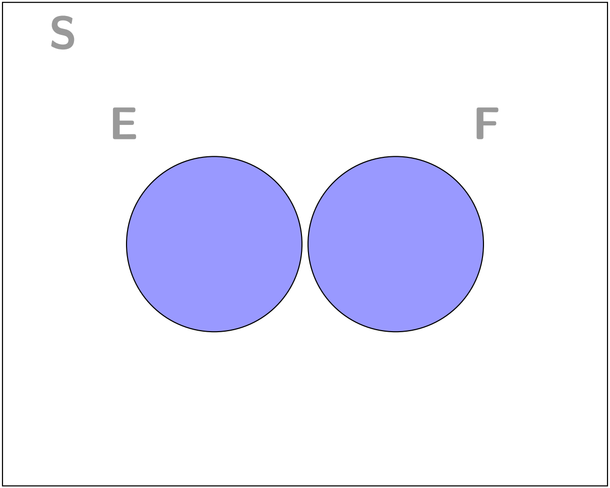



**Definition**

Two events $E$ and $F$ are mutually exclusive (i.e. disjoint) if $E \cap F = \emptyset$.




- No overlap between two events

# Example of Two Dice Rolled

- $E =$ sum is odd
- $F =$ sum is 4

# Example of Two Dice Rolled

- $E$ = sum is odd
- $F$ = sum is 4

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|---|---|---|---|---|---|---|
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| 1st Dice | (3, 1) | (3, 2) | (3, 3) | (3, 4) | (3, 5) | (3, 6) |
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| | (6, 1) | (6, 2) | (6, 3) | (6, 4) | (6, 5) | (6, 6) |

# Let's Combine Them!

- Naturally, we can combine notation together to represent certain spaces/subsets
  - e.g. $E \cup F^C$
  - e.g. $(E \cap F^C) \cup (E^C \cap F)$
  - e.g. $S^C$

# Union and Intersection, Generalized

We can define the union and intersection for events $E_1, \ldots, E_n$:

- Union: $\bigcup_{i=1}^{n} E_i$
- Intersection: $\bigcap_{i=1}^{n} E_i$

For countably infinite events, we can replace $n$ with $\infty$.

# Set Operations

- Communative (flip order)
  - $E \cup F = F \cup E$
  - $E \cap F = F \cap E$
- Associative (groupings)
  - $E \cup (F \cup G) = (E \cup F) \cup G$
  - $E \cap (F \cap G) = (E \cap F) \cap G$
- Distributive (apply within groupings)
  - $E \cup (F \cap G) = (E \cup F) \cap (E \cup G)$
  - $E \cap (F \cup G) = (E \cap F) \cup (E \cap G)$

# Proofs

- We will be writing proofs periodically throughout this course
  - Builds intuition about material
  - Formal mathematical language
- We will slowly go through the steps in the coming lectures

# Set Equality

To prove that two sets A and B are equal to each other (i.e. $A = B$), we need to show containment in both directions, i.e.

$$A = B \iff A \subseteq B \text{ and } B \subseteq A$$

- $A \implies B$: if $A$, then $B$
- $A \iff B$: if and only if; if $A$, then $B$, and if $B$, then $A$

# DeMorgan's Laws

For two events $E_1$ and $E_2$,

1. $(E_1 \cup E_2)^C = E_1^C \cap E_2^C$
2. $(E_1 \cap E_2)^C = E_1^C \cup E_2^C$

# DeMorgan's Laws

For two events $E_1$ and $E_2$, suppose $E_1$ is a sophomore in this classroom and $E_2$ is a math/stats major. Then,

1. $(E_1 \cup E_2)^C = E_1^C \cap E_2^C$
   - "Those that aren't sophomore or math/stats majors" are the same as "those that aren't sophomore and those that aren't a math/stats major".
2. $(E_1 \cap E_2)^C = E_1^C \cup E_2^C$
   - "Those that aren't sophomore and math/stats majors" are the same as "those that aren't sophomore or those that aren't a math/stats major".

To prove the first law:

$$(E_1 \cup E_2)^C = E_1^C \cap E_2^C$$

## Proof of First Law

We need to show set containment in both directions:

1. $(E_1 \cup E_2)^C \subseteq E_1^C \cap E_2^C$
2. $E_1^C \cap E_2^C \subseteq (E_1 \cup E_2)^C$

$$(E_1 \cup E_2)^C = E_1^C \cap E_2^C$$

## Proof of First Law (cont.)

We first show $(E_1 \cup E_2)^C \subseteq E_1^C \cap E_2^C$. Let $x$ be an element in $(E_1 \cup E_2)^C$. Then,

$$\begin{aligned}
& x \in (E_1 \cup E_2)^C \\
\Longrightarrow\ & x \notin E_1 \cup E_2 && \text{by def. of complement} \\
\Longrightarrow\ & x \notin E_1 \text{ and } x \notin E_2 && \text{by def. of union} \\
\Longrightarrow\ & x \in E_1^C \text{ and } x \in E_2^C && \text{by def. of complement} \\
\Longrightarrow\ & x \in E_1^C \cap E_2^C && \text{by def. of intersection}
\end{aligned}$$

Since $x \in E_1^C \cap E_2^C$, then $(E_1 \cup E_2)^C \subseteq E_1^C \cap E_2^C$.

$$(E_1 \cup E_2)^C = E_1^C \cap E_2^C$$

## Proof of First Law (cont.)

We now show $E_1^C \cap E_2^C \subseteq (E_1 \cup E_2)^C$. Let $x$ be an element in $E_1^C \cap E_2^C$. Then,

$$\begin{aligned}
& x \in E_1^C \cap E_2^C \\
\implies & x \in E_1^C \text{ and } x \in E_2^C && \text{def. of intersection} \\
\implies & x \notin E_1 \text{ and } x \notin E_2 && \text{by def. of complement} \\
\implies & x \notin E_1 \cup E_2 && \text{by def. of union} \\
\implies & x \in (E_1 \cup E_2)^C && \text{by def. of complement}
\end{aligned}$$

Since $x \in (E_1 \cup E_2)^C$, then $E_1^C \cap E_2^C \subseteq (E_1 \cup E_2)^C$.
Since $(E_1 \cup E_2)^C \subseteq E_1^C \cap E_2^C$ and $E_1^C \cap E_2^C \subseteq (E_1 \cup E_2)^C$, then $(E_1 \cup E_2)^C = E_1^C \cap E_2^C$. □

# On Your Own Time

Prove DeMorgan's second law, using techiques seen in the proof of the first law.

# DeMorgan's Laws, Generalized

DeMorgan's laws can be extended to a finite number of events $n$, as well as countably infinite number of events. For a finite number of events:

1. $(\bigcup_{i=1}^{n} E_i)^C = \bigcap_{i=1}^{n} E_i^C$
2. $(\bigcap_{i=1}^{n} E_i)^C = \bigcup_{i=1}^{n} E_i^C$

# Naive Definition of Probability

Suppose all outcomes of an experiment are equally-likely, then the probability of an event $A$ is given by:

$$P(A) = \frac{\text{number of outcomes in A}}{\text{number of outcomes in S}}$$

- Definition is helpful, but not widely applicable
  - Why?
- Next class: axiomatic definition of probability

# See You Tuesday!

- Homework #0 is due on Friday, 9/4 at 11:59pm
  - Includes survey, setup of tech, and turning in quick problem
- Read through Chapter 1 of Introduction to Probability
- Office hours: none on Monday, Tuesday from 3-4:30pm in Rocky 106A